

Internal energy

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 4 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	Define internal energy.	The total energy stored by the particles that make up a system.	☐
2 ★	State how internal energy is related to the energies of the particles.	Internal energy = total kinetic energy + total potential energy of the particles	☐
3 ★	State which part of the internal energy changes when the temperature changes.	The kinetic energy of the particles	☐
4	Describe how the average kinetic energy of the particles is related to temperature.	The greater the average kinetic energy, the higher the temperature.	☐
5	State which part of the internal energy changes when a substance changes state.	The potential energy of the particles	☐
6	The potential energy of the particles changes at a constant temperature. State what happens to the substance.	It changes state.	☐
7	State the two things that can happen when a substance is heated.	Its temperature can increase, or it can change state.	☐
8 ★	Describe the change in temperature of a substance during a change of state.	The temperature remains constant.	☐
9	State the two things that can happen when a substance is cooled.	Its temperature can decrease, or it can change state.	☐
10	In which state of matter is the internal energy of a substance highest?	Gas	☐
11	State what heating does to the internal energy of a system.	It increases it (the particles gain energy).	☐
12	State three things that the temperature rise of a heated substance depends on.	Its mass, its specific heat capacity and the energy transferred to it	☐